

Abstract No.

The spatial distribution of tryptophan metabolic fields promote niche specific immunosuppressive programs in gliomas

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Tryptophan (Trp) metabolism leads to the production of neurotransmitters, protective and neurotoxic metabolites. Trp catabolizing enzymes (TCEs) produce a plethora of metabolites that drive immune suppression and cell motility. In particular, indoleamine-2,3-dioxygenase 1 (IDO1) and tryptophan-2,3-dioxygenase 2 (TDO2) which produce kynurenine (Kyn), and interleukin-4 induced 1 (IL4I1), which generates indole-3-pyruvate (I3P), mediate these immune suppressive effects by activating the aryl hydrocarbon receptor (AHR). Glioma cells strictly select one main TCE for catabolizing Trp, with extreme rare events showing more than one TCE expressed in the same cell. Moreover, TCEs are found to be dominating certain glioma and microenvironment cell types. Analyzing spatial data showed that TCEs, although sparsely expressed, follow a clear spatial distribution leading to niche specific immunosuppressive programs. Our AI models demonstrate that these effects are due to specific metabolite distribution patterns, specifically those of kynurenine (Kyn) and indole-3-pyruvate (I3P). We applied vision transformer AI models to map these spatial metabolic fields and we were able to use HE slide sections to classify patient tumors based on high and low Trp metabolic activity. We show that glioma patient tumors with high Trp metabolic activity fuels an immunosuppressive microenvironment only upon immunotherapy and not upon standard of care. The high Trp active patients had a four times higher hazard and worse outcome. Our AI models provide a proof of concept for streamlined integration of Trp activity in standard clinical routine. Moreover, our models will improve the stratification of glioma patients to immunotherapies.



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